

Econ 30065: Microeconomics

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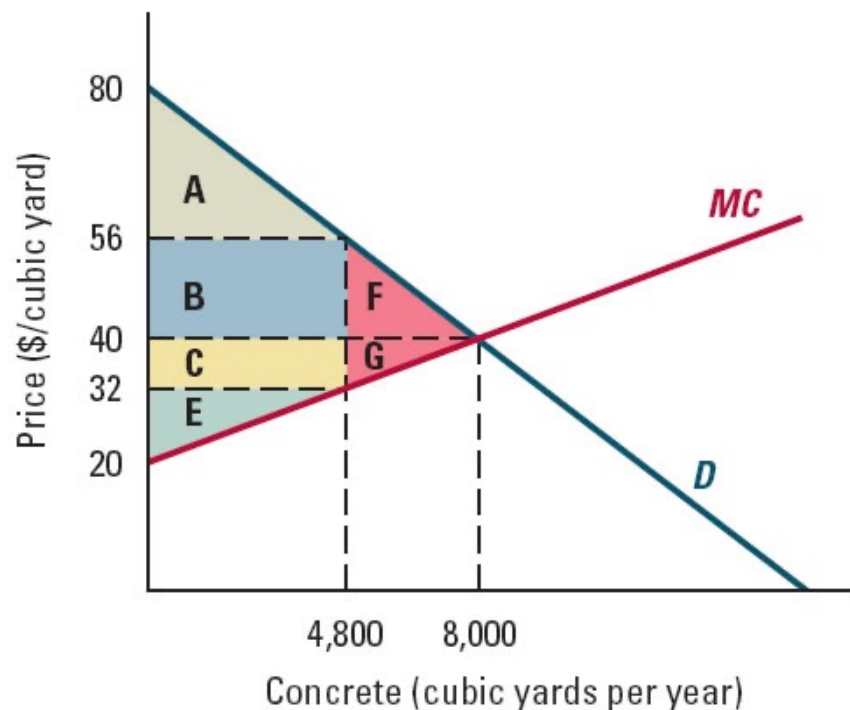
Pricing Policies

Topics

- Alternative pricing strategies for a Monopolist:
 - Price discrimination: pricing to extract surplus
 - Perfect price discrimination/two-part tariff
 - Price discrimination based on observable customer characteristics

Monopolist's problem

- We saw how a monopolist sets the price (and quantity) to maximize his profits, if he can charge **only one price** for every unit
- He could increase profits if he could:
 - Take area A (consumer surplus) charging different prices for different units
 - Take area F+G charging less for those units of the good that are not sold but for which willingness to pay is higher than MC



Price discrimination

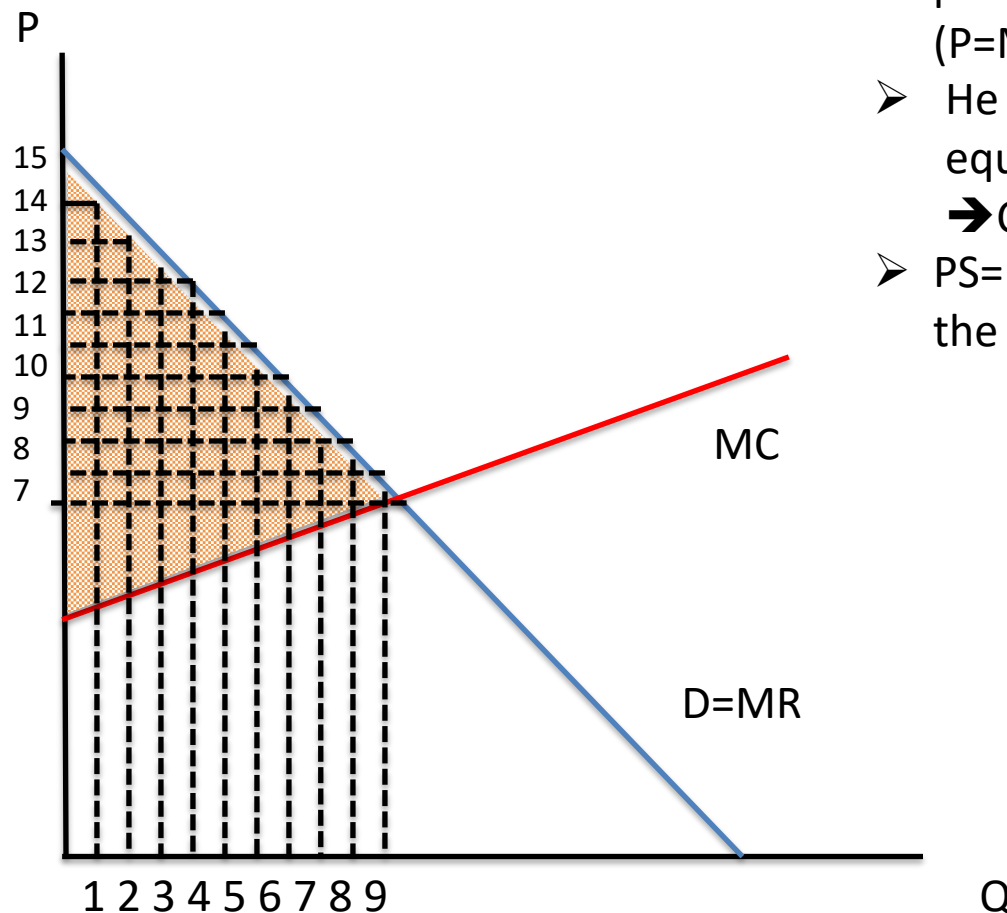
- Monopolist might be able to do better by ***price discriminating***: charging different prices for different units of the same good
- To be able to price discriminate:
 1. A firm must have some market power
 - If not, a price above marginal cost will result in zero sales
 2. The good or service must be difficult to resell
 - Otherwise few sales will occur at the higher price
 3. Firm must also be able to distinguish sales for which the purchasers have a high willingness to pay from those they have a low willingness to pay:
 - Perfect price discrimination
 - Perfect price discrimination
 - Two-part tariff
 - Price discrimination based on observable customer characteristics

Perfect price discrimination

A monopolist can engage in **perfect price discrimination** if:

- He knows a customer's willingness to pay for each unit he sells
- He can charge a different price for each unit
- ➔ He will charge each customer exactly his willingness to pay for the good! $P_i = WTP_i$
- ➔ He will sell all the units for which willingness to pay (marginal revenue) is higher (or equal) than marginal cost: he will stop at $P = MR = MC$!

Perfect price discrimination



- WELFARE EFFECT: EFFICIENCY:
- Monopolist will sell same Q as in perfect competition
($P=MR=MC$) → no DWL!
- He charges each customer a price equal to his willingness to pay
→ $CS=0$
- PS = orange area (he captures all the surplus!)

Example

Suppose inverse market demand is given by $P=50-Q$ and that the monopolist can perfectly price discriminate among consumers. Assume moreover it is illegal to resell units of the good. Monopolist total costs are given by $TC=25+3Q$. What is the Q that the monopolist will optimally sell? What about consumer surplus, DWL and monopolist profits?

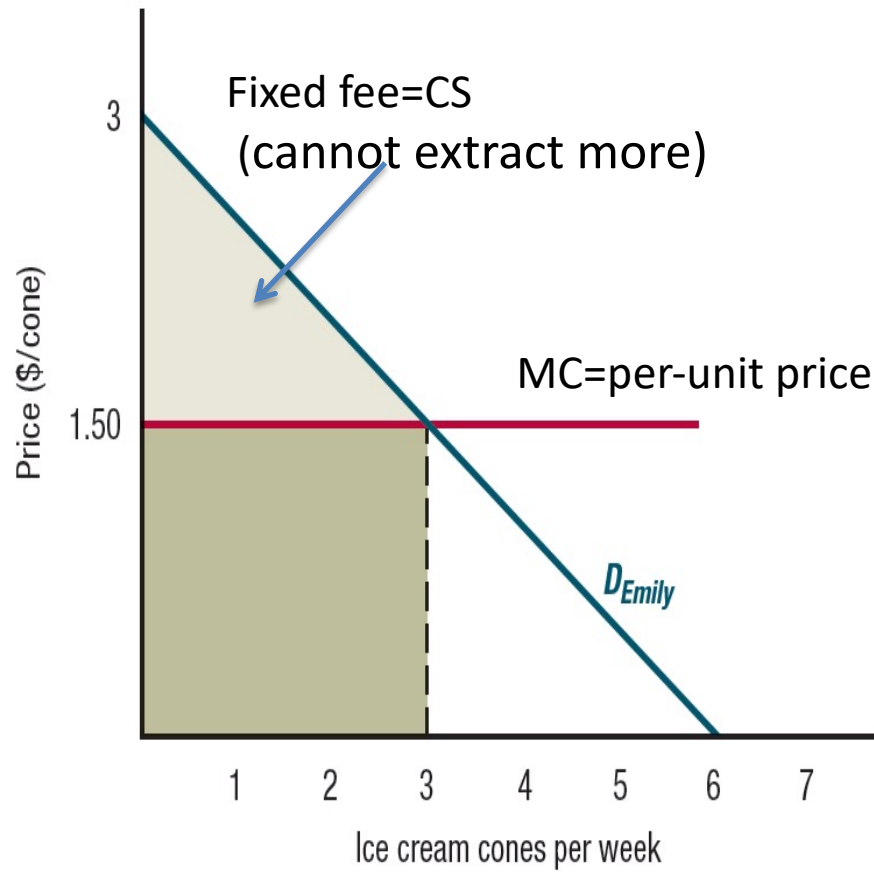
Solution

- Set $P=MC$ to find the quantity that the monopolist will sell:
 $50-Q=3 \rightarrow Q^M=47$, the monopolist will set prices ranging from $P=50$ to $P=3$.
- $CS=0$
- $DWL=0$
- $PS=(50-3)47/2=1104.5$
- Monopolist profits
 - SURPLUS-FIXED COSTS :
 - $1104.5-25=1079.5$
 - (or $TR-TC$)
 - $TR=\text{area of trapezoid: } 0.5(50+3)47=1245.5$
 - $TC=25+3 \times 47=166$
 - $\text{Profits}=1245.5-166=1079.5$

Two-Part Tariffs

- In practice, it is very difficult to implement perfect price discrimination based on each consumer's WTP for every different level of quantity demanded
- Two-part tariffs are another (simpler) pricing plan that allows a perfectly discriminating monopolist to maximize profit
- With a ***two-part tariff***, consumers pay an initial fixed fee plus a separate per-unit price for each unit they buy
 - Examples: amusement parks, rental car companies, nightclubs
- Advantage is simplicity: name just two prices for each buyer
- To maximize profit:
 - set per-unit price equal to marginal cost
 - Fixed fee=CS left after the per unit price

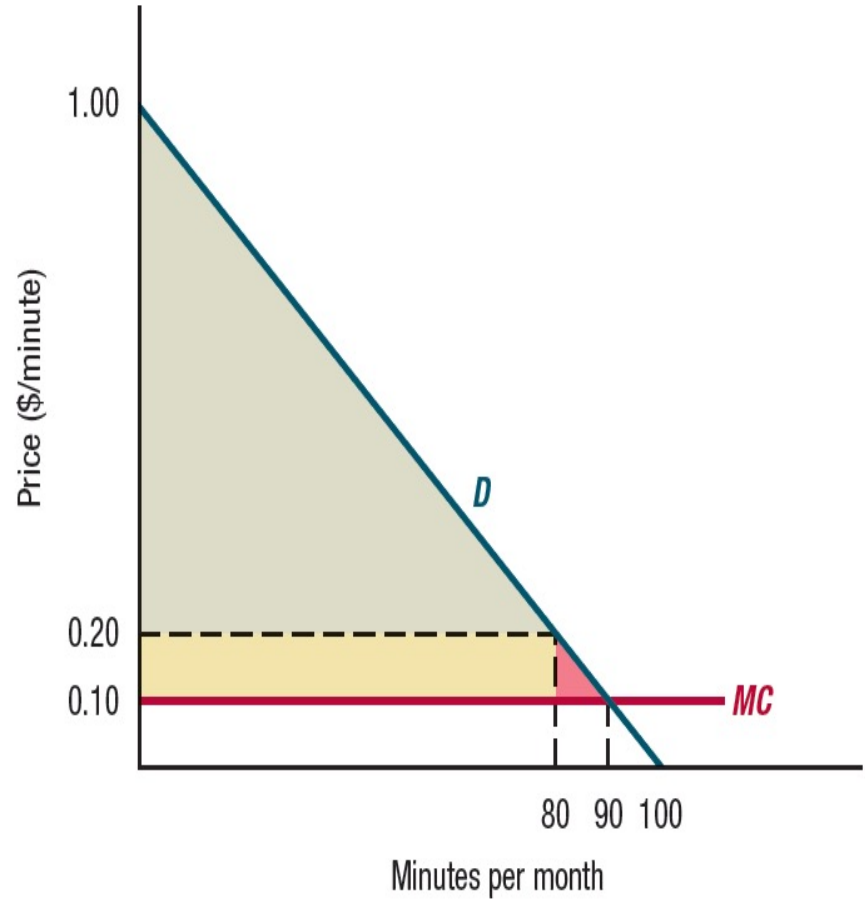
Two-Part Tariffs



- WELFARE EFFECT: EFFICIENCY:
- Monopolist will sell same Q as in perfect competition ($P=MC$) → no DWL!
- He extracts all the CS → $CS=0$
- Same result as with perfect price discrimination!

Profit with a Two-Part Tariff and Identical Consumers

By lowering its per-unit charge to its marginal cost, and raising its fixed fee, the firm can increase its profit to the maximum possible



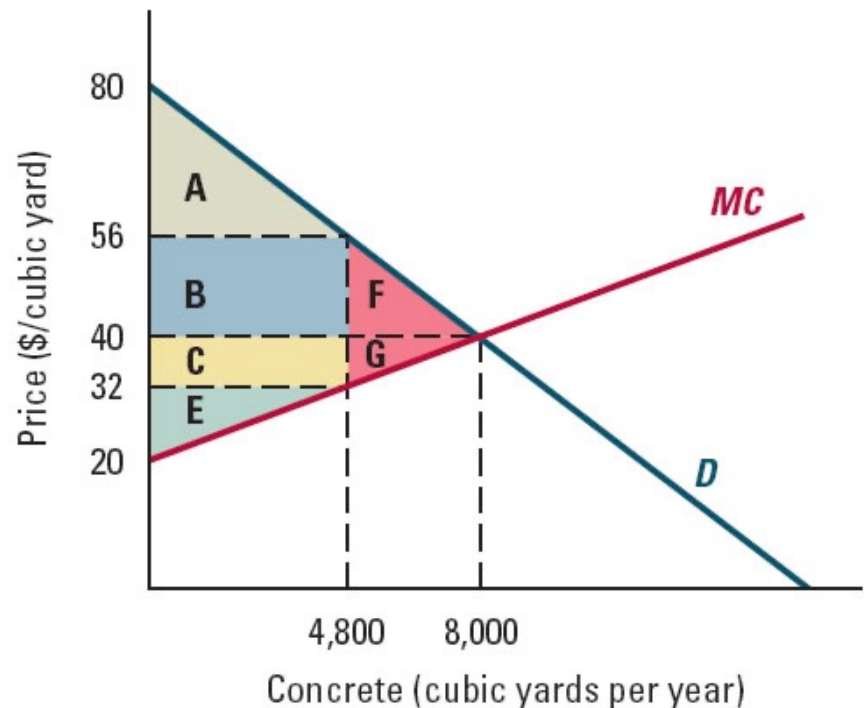
Example

If the quantity demanded by a buyer is $Q^d=63-P$ and TC for the firm is $10+ 10Q^2$, what is the two parts tariff that maximizes its profits?

- 1) Set $P=MC$: $63-Q=20Q \rightarrow Q=3, P=60$ (per unit price)
- 2) Compute CS: $0.5(63-60)3=4.5$ =fixed fee
- 3) $PS=63 \times 0.5 \times 3=94.5$
- 4) Profits: $TR-TC=4.5+60 \times 3-10-10 \times 9=84.5$

Perfect price discrimination-summing up

With perfect price discrimination (or 2 part tariff) the monopolist can take both A and G+F



Observable Customer Characteristics

- Most often a firm's ability to price discriminate is imperfect
- May be able to sort consumers into rough groups based on **observable characteristics**: a firm can distinguish, even if imperfectly, consumers with a high versus low willingness to pay
 - Example: small town movie theater with four consumer groups (adults, seniors, students, kids)
- To maximize profit consider each group's demand curve separately
 - Set price to maximize profit earned from that group ($MR=MC$ for each group) → find different prices for different groups!

Observable Customer Characteristics

- Charge a higher price to groups with less elastic demand:

$$\frac{P - MC}{P} = -\frac{1}{E^D}$$

- Profits higher than without price discrimination (but not as high as with perfect price discrimination)
- $DWL > 0$ (inefficiency)
- $CS > 0$

Example

Suppose the demand functions for movie tickets for students vs other adults are given by:

$$Q_S^d = 800 - 100P$$

$$Q_A^d = 1600 - 100P$$

and that the monopolist MC is constant at 2\$.

To maximize profits: $MR_S = MC$ and $MR_A = MC$

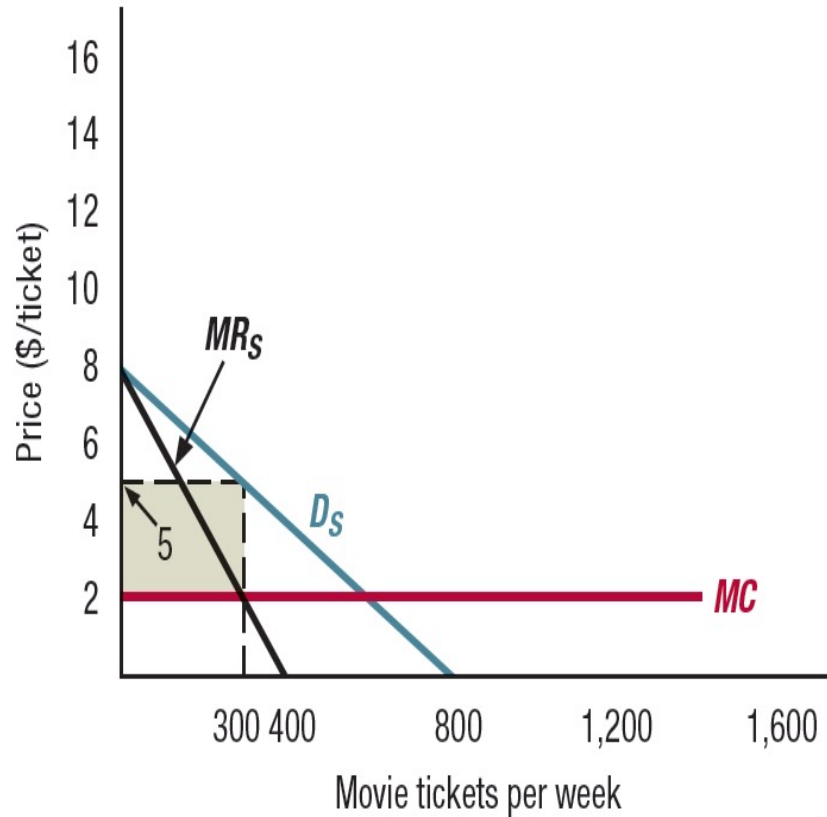
$$8 - Q_S/50 = 2 \rightarrow Q_S = 300, P_S = 5$$

$$16 - Q_A/50 = 2 \rightarrow Q_A = 700, P_A = 9$$

$$\text{Profits} = 300 \times 5 + 700 \times 9 - 2 \times 1000 = 5800$$

Example

(a) College students



(b) Other adults

